

# Ephemeral Meta-Weights: Turning Transformer FFN Parameters into Runtime-Generated Operands

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## Abstract

Transformer scaling is usually framed as a choice between storing more parameters, quantizing them more aggressively, or activating fewer of them. We study a different axis: *parameter lifetime*. A feed-forward network (FFN) matrix need not always be a durable tensor that is fetched, cached, differentiated, and synchronized as model state. It can instead be represented, for selected calls, by a short-lived runtime operand generated from context and consumed by a kernel. We study this idea through MWG-EW, an Ephemeral Meta-Weight architecture for Transformer FFNs.

MWG-EW replaces selected persistent FFN matrices with conditional rank- $r$  descriptors. A generator emits factors for the current layer and token group; an associative  $(XU_r)V_r$  schedule consumes them under a proposed no-write-back lifecycle. This reframes efficient Transformers around a contract between learning and systems: the model learns *when* a generated descriptor is useful, while the runtime must keep that descriptor from becoming persistent bandwidth traffic again.

The paper contributes a parameter-lifetime abstraction, an ephemeral FFN design, scaling laws for descriptor traffic and distributed gradient communication, a benefit-supervised dense-fallback router, and a reproducible Ascend NPU evidence trail. The strongest new systems evidence is a fresh three-repeat, all-eight-NPU ASI3 Ascend 910B2 sweep over ranks 32–512. At rank 128, MWG-EW reduces estimated descriptor bytes by 24.9x and trainable FFN state by 18.4x while improving measured training throughput from 1.483M to 2.428M tokens/s, a  $1.638 \pm 0.030x$  training speedup. Across all tested ranks, mean training speedup remains positive (1.538x–1.646x) and estimated descriptor traffic falls by 6.2x–99.6x. The forward path is not faster in the unfused prototype, so the validated systems claim is deliberately scoped to training, state, and distributed communication.

Checkpoint-backed Qwen2.5-1.5B experiments identify the key algorithmic result: MWG-EW is strongest as a selective FFN replacement rather than a blanket one. A fresh ASI3 recovery run on layer 16 further tightens this picture: at rank 384, the distilled generator reaches relative MSE 0.101849 and cosine 0.947711, far ahead of static SVD at 0.836682 and 0.404468 for the same rank. A 2026-06-11 two-NPU ASI3 recovery independently preserves the result at relative MSE 0.130428 and cosine 0.932599, versus static SVD at 0.835157 and 0.406409. The same checkpoint family improves a calibration held-out slice from dense PPL 3.748 to 2.628, and corrected five-suite validation remains positive in aggregate: 0.9352x token-weighted perplexity across 74,456 tokens for the always-patched layer, and 0.8493x across 39,394 tokens for the corrected hybrid manifest. Selective dense-fallback routing gives 0.969x, 0.955x, and 0.940x at 3.59%, 6.69%, and 14.71% actual patched tokens. A fresh 2026-06-24 three-seed reproduction is more conservative: the same checkpoint is strongly positive in-distribution (0.404x calibration slice over 35,374 tokens) and on GSM8K (0.965x), but under leakage-clean seeded splits the always-patched five-suite aggregate is above dense (1.409x), so a uniformly safe five-suite always-patched policy is not yet established. The current validated claim is therefore positive but bounded: generated conditional descriptors, learned eligibility selection, and a proposed no-write-back lifecycle make FFN parameter movement a useful architectural variable.

# 1 Introduction

Modern Transformers are built around a durable-parameter assumption: large weight tensors are stored as model state, fetched whenever a layer executes, updated by the optimizer, and synchronized during distributed training. This assumption is natural for learning, but it is not free for systems. In many serving and training regimes, FFN projections are among the largest repeatedly streamed objects in the block. Quantization, sparsity, expert routing, and kernel engineering reduce different pieces of this cost, but they typically preserve the same lifetime: a parameter tensor remains a durable object that the runtime must move.

This paper asks whether parameter lifetime itself can become an architectural choice. Instead of asking only how many parameters a layer stores, we ask how long a weight-like object must exist and where it is allowed to live. The answer is a new execution primitive: an *ephemeral meta-weight*. An ephemeral meta-weight is a generated descriptor that is created from context, used as a kernel operand, and then released or overwritten. It is not a compressed checkpoint tensor and it is not a persistent adapter. Its value lies in changing the movement pattern of parameters: fewer durable bytes are fetched from memory and fewer durable gradients are synchronized across workers.

We instantiate this idea in Transformer FFNs with MWG-EW. A compact generator emits rank- $r$  factors for selected up, gate, and down projections. The FFN consumes those factors with the associative schedule  $(XU_r)V_r$ . The runtime contract is deliberately strong: generated descriptors should be consumed at the closest available scratch/cache level and should not be written back as durable model weights. The learning problem is also deliberately strong: the model must learn not only a useful low-rank approximation, but also an eligibility policy deciding when the generated path is worth using.

This is different from three familiar efficient-model directions. Hypernetworks can generate weights, but commonly materialize them as ordinary tensors. Low-rank adapters store their factors persistently. IO-aware attention kernels such as FlashAttention optimize activation movement, while attention-side KV compression methods such as MLA compress past-token state. MWG-EW targets FFN weight movement and makes the generated factor’s lifetime part of the method.

The present evidence leads to a clear claim. A naive always-patched low-rank replacement is not the right operating mode. However, once the patch is treated as a benefit-supervised conditional operand rather than a universal replacement, a robust positive regime appears: a deployable router can patch a limited fraction of tokens while preserving or improving broad perplexity. MWG-EW is therefore valuable not as a full dense-model substitute, but as a new selective execution primitive with measurable quality and systems benefits.

The paper is written around falsifiable claim boundaries. We distinguish measured training throughput from estimated descriptor traffic, full-call systems savings from routed end-to-end savings, and quality-preserving selective replacement from always-patched replacement. This separation is important for the contribution: the method’s novelty is not a single benchmark win, but a new parameter-lifetime interface that exposes when generated FFN operands can replace durable weights without degrading model behavior.

**Definition 1** (Ephemeral meta-weight). *An ephemeral meta-weight for a layer is a descriptor  $D_t = G_\theta(c_t, \ell)$  that is generated from the current execution context, consumed by the layer’s kernel, excluded from the persistent model-state interface, and released or overwritten before it becomes durable off-chip parameter state.*

**Definition 2** (Parameter-lifetime class). *A weight-like object has lifetime class  $(\tau, \sigma)$ , where  $\tau$  is its maximum reuse interval and  $\sigma$  is the deepest memory or synchronization boundary it crosses. Dense FFN weights have long  $\tau$  and cross the model-state and gradient-synchronization boundary.*

MWG-EW aims for short  $\tau$  and a scratch/cache-level  $\sigma$  for descriptors, while keeping only generator and basis-bank state durable.

**Definition 3** (Eligibility-preserving replacement). *Let  $g(x) \in \{0, 1\}$  decide whether a token or token group uses the ephemeral path. A replacement is eligibility preserving at budget  $\mathcal{B}$  if  $\mathbb{E}[g(x)] \leq \mathcal{B}$  and its evaluated task loss is no worse than the dense baseline within the stated confidence or seed envelope. This separates the quality claim from broad always-patched replacement.*

**Contributions.** We make the following contributions:

- **Parameter lifetime as an architectural variable.** We formalize ephemeral meta-weights and distinguish them from compressed checkpoints, persistent adapters, and expert weights.
- **A Transformer FFN design.** We give a generator-conditioned FFN that emits conditional rank- $r$  descriptors for up, gate, and down projections and consumes them through an associative schedule.
- **A systems contract.** We specify a no-write-back descriptor lifecycle and derive scaling laws for descriptor bytes, working set, memory-bound latency, and distributed gradient communication.
- **Benefit-supervised eligibility routing.** We evaluate a deployable dense-fallback router trained from dense-vs-patched token loss deltas but executed from pre-FFN hidden-state features.
- **A scoped empirical case.** We report eight-accelerator Ascend NPU systems evidence and checkpoint-backed Qwen2.5-1.5B quality controls showing a stable positive selective-routing regime at target patch fractions 0.05 and 0.10, while explicitly marking always-patched and hardware-counter claims as unvalidated.

## 2 Related Work

**Transformer FFNs and parameter efficiency.** The Transformer architecture [18] places gated FFN blocks between attention sublayers, and these blocks dominate parameter count and weight-traffic in modern language models. Most efficiency work on these blocks falls into four families: conditional computation, low-rank or structured parameterization, weight generation, and IO-aware kernels. We position MWG-EW against each in turn.

**Conditional computation and experts.** Mixture-of-Experts (MoE) layers reduce activated compute by routing each token to a small subset of expert FFNs [17], and modern large-scale systems such as GShard [12], Switch Transformers [7], and DeepSeek-V2 [5] scale this idea with learned routing and sharding. Adaptive computation time similarly learns how much computation to spend per input [9]. MWG-EW shares the conditional spirit of these methods, but it conditions on a different object: rather than selecting among persistent expert weights, it decides whether a weight-like operand should be generated and kept ephemeral at all. Our benefit-supervised dense-fallback router is closer to a budgeted eligibility gate than to a load-balanced expert assignment.

Table 1: Evidence map for the paper’s main claims. This table separates what is directly measured, what is analytically estimated from measured geometry, and what remains an implementation target.

| Claim                                                                                       | Evidence in this paper                                                                                                                                                | Status                                                                                                      |
|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Training-side FFN replacement can reduce state and communication while improving throughput | Three-repeat ASI3 eight-NPU sweep over ranks 32–512; ASI1 and ASI2 reliability runs                                                                                   | Measured and replicated directionally                                                                       |
| Generated descriptors reduce full-call FFN operand bytes                                    | Rank- $r$ descriptor analysis and measured FFN geometry                                                                                                               | Analytically estimated                                                                                      |
| Selective routing preserves or improves broad perplexity at small patch fractions           | Leakage-clean broad seed sweeps, leave-suite-out sweeps, and a fresh ASI3 router/global split with actual mixed forwards                                              | Measured for three suite families                                                                           |
| Always-patched replacement is safe in-distribution and on GSM8K                             | 2026-06-11 single-run calibrated ASI3 five-suite validation is below dense (0.9352x), but the 2026-06-24 three-seed leakage-clean replication is above dense (1.409x) | Split: supported in-distribution and on GSM8K; <i>not</i> established as uniformly safe under seeded splits |
| No-write-back descriptor lifecycle avoids off-chip descriptor write-back                    | Runtime contract and operator probe; no fused-like operator observed                                                                                                  | Implementation target                                                                                       |

**Low-rank and structured weights.** Low-rank factorization of linear layers is a classic way to shrink parameter count and compute [16], and structured matrix families such as Monarch matrices generalize this with hardware-efficient parameterizations [3]. LoRA and QLoRA add low-rank factors for parameter-efficient adaptation [11, 6]. These approaches all keep the low-rank factors as durable, stored parameters. MWG-EW instead treats the factors as conditionally generated operands with a short lifetime, so the saving is in parameter *movement* and synchronization rather than only in stored count.

**Weight generation and hypernetworks.** Hypernetworks generate the weights of another network from a learned embedding [10], and one-shot architecture search systems such as SMASH predict weights to evaluate candidate models cheaply [2]. The key difference here is lifetime and runtime contract: prior weight-generation methods materialize ordinary tensors, whereas MWG-EW specifies a no-write-back lifecycle so generated descriptors are consumed at the closest scratch/cache level and never recommitted as durable off-chip parameter state.

**Post-training compression and quantization.** GPTQ [8] and AWQ [13] compress LLM weights by reducing bits per parameter while preserving the durable-tensor lifetime. MWG-EW is orthogonal and composable: it changes when and where a weight-like object exists, not only its precision, and a generated descriptor could itself be quantized.

**Communication-efficient distributed training.** Gradient compression methods reduce all-reduce traffic by sparsification [14], quantization [1], or low-rank projection of gradients [19], while

optimizer-state sharding such as ZeRO reduces per-worker memory [15]. MWG-EW reduces the size of the synchronized object itself: only generator and basis-bank parameters are durable, so the dominant FFN gradient-byte term is replaced by the much smaller  $6r(d + m)$  term. These techniques are complementary and can be applied on top of the remaining generator parameters.

**IO-aware kernels and KV-cache compression.** FlashAttention reduces attention activation movement through IO-aware kernels [4], and MLA-style methods compress the attention KV cache [5]. Both target activation- or past-token state, whereas MWG-EW targets FFN weight movement and is orthogonal to them. Table 3 summarizes the resulting positioning: the distinguishing axis is parameter lifetime, not the specific approximation family.

### 3 Architecture

#### 3.1 Baseline Gated FFN

A gated Transformer FFN maps hidden states  $X \in \mathbb{R}^{B \times S \times d}$  to

$$Y = (\phi(XW_{\text{gate}}) \odot XW_{\text{up}}) W_{\text{down}}, \quad (1)$$

where  $W_{\text{up}}, W_{\text{gate}} \in \mathbb{R}^{d \times m}$ ,  $W_{\text{down}} \in \mathbb{R}^{m \times d}$ , and  $\phi$  is typically SiLU. With fp16 weights, one block stores approximately  $6dm$  bytes when  $m$  dominates  $d$ .

#### 3.2 Ephemeral Meta-Weight FFN

MWG-EW replaces each projection matrix by a conditional descriptor pair:

$$(U_r^{(p)}, V_r^{(p)}) = G_\theta(c_t, \ell, p), \quad XW_p \approx (XU_r^{(p)})V_r^{(p)}, \quad (2)$$

where  $p \in \{\text{up}, \text{gate}, \text{down}\}$ ,  $c_t$  is a token-group context,  $\ell$  is the layer id,  $U_r \in \mathbb{R}^{d_{\text{in}} \times r}$ , and  $V_r \in \mathbb{R}^{r \times d_{\text{out}}}$ . The FFN becomes

$$\tilde{Y} = \left( \phi((XU_r^{(g)})V_r^{(g)}) \odot (XU_r^{(u)})V_r^{(u)}) \right) U_r^{(d)}V_r^{(d)}. \quad (3)$$

We use a basis-bank generator:

$$\alpha = \text{softmax}(A_\ell h_\theta(c_t)), \quad (4)$$

$$U_r^{(p)} = \sum_{k=1}^K \alpha_k U_k^{(p)} + \Delta U_\theta^{(p)}(c_t, \ell), \quad (5)$$

$$V_r^{(p)} = \sum_{k=1}^K \alpha_k V_k^{(p)} + \Delta V_\theta^{(p)}(c_t, \ell). \quad (6)$$

The bank gives stable reusable directions; the residual head gives input-conditioned adaptation. Practical kernels may generate only rank-channel scales or tile-local residuals when full descriptor synthesis is too expensive.

### 3.3 Descriptor Lifecycle

The runtime contract is simple and testable:

1. Read token context  $c_t$  and layer id  $\ell$ .
2. Generate descriptor factors  $(U_r, V_r)$  for each FFN projection.
3. Store descriptor tiles in an on-chip scratch region when the target accelerator exposes such memory; otherwise keep the same volatile lifecycle at the closest available cache level.
4. Compute each projection with the associative schedule  $(XU_r)V_r$ .
5. Release or overwrite each scratch tile before the next token group.
6. Synchronize gradients only for generator and basis-bank parameters.

The important distinction is not only low rank. A low-rank factor stored as a long-lived model tensor still creates persistent parameter traffic. The ephemeral contract allows a runtime to avoid writing generated descriptors back to off-chip memory and allows distributed training to synchronize only the generator-side state.

## 4 Analysis

### 4.1 Traffic Scaling

For a gated FFN with three projections and fp16 elements, dense weight traffic per block is

$$\mathcal{T}_{\text{dense}} = 6dm. \quad (7)$$

Ignoring the small generator context, rank- $r$  descriptor traffic is

$$\mathcal{T}_{\text{mwg}} = 2r(d + m) + 2r(d + m) + 2r(m + d) = 6r(d + m). \quad (8)$$

The idealized traffic ratio is therefore

$$\Gamma(r) = \frac{\mathcal{T}_{\text{mwg}}}{\mathcal{T}_{\text{dense}}} = \frac{r(d + m)}{dm}. \quad (9)$$

At  $d = 4096$ ,  $m = 14336$ , and  $r = 128$ , this gives  $\Gamma = 0.0402$ , or a 24.9x reduction in descriptor bytes.

**Proposition 1** (Memory-bound speedup condition). *Assume per-block latency can be approximated by  $L = L_0 + \kappa\mathcal{T}$  with fixed launch/compute term  $L_0$  and bandwidth coefficient  $\kappa > 0$ . Then MWG-EW improves latency whenever  $\mathcal{T}_{\text{mwg}} < \mathcal{T}_{\text{dense}}$ , and the asymptotic speedup approaches  $1/\Gamma(r)$  as  $L_0$  becomes small relative to memory traffic.*

### 4.2 Gradient Communication

In data-parallel training, dense FFN gradients require all-reduce traffic proportional to  $6dm$  bytes per block. MWG-EW synchronizes generator and descriptor-bank parameters, whose leading term is  $6r(d + m)$  plus the small MLP state. The same ratio  $\Gamma(r)$  controls the dominant gradient-byte reduction. This is orthogonal to optimizer sharding [15] and tensor parallelism: those systems can still be used on the remaining generator parameters. It is also complementary to gradient-compression methods that sparsify [14], quantize [1], or low-rank-project [19] the gradients of a fixed parameter set; here the synchronized parameter set itself is smaller, so the two families of savings multiply rather than overlap.

### 4.3 Computational Complexity

Let  $T = BS$  be the number of tokens processed by one FFN call and let  $q$  be the token-group reuse factor for a generated descriptor. The dense gated FFN performs three large matrix products and has leading compute

$$\mathcal{C}_{\text{dense}} = \Theta(3Tdm) \quad (10)$$

multiply-accumulate operations. The associative MWG-EW path computes  $XU_r$  and then  $ZV_r$  for the up, gate, and down projections:

$$\mathcal{C}_{\text{mwg,assoc}} = \Theta(3Tr(d+m)). \quad (11)$$

If a basis bank with  $K$  entries is explicitly blended once per token group, the descriptor-generation term is

$$\mathcal{C}_{\text{gen}} = \Theta(3 \lceil T/q \rceil Kr(d+m) + \lceil T/q \rceil C_G), \quad (12)$$

where  $C_G$  is the small context-network cost. Kernels that generate only rank-channel scales or tile-local residuals replace the  $Kr(d+m)$  term by the corresponding tile size. Thus the leading compute ratio, ignoring amortized generation, is again  $\Gamma(r) = r(d+m)/(dm)$ . The method is attractive when  $r \ll \min(d, m)$  and  $q$  is large enough to amortize generation; it can be unattractive when descriptors are regenerated for very small token groups or when unfused factor matmuls lose accelerator utilization.

### 4.4 Working-Set Complexity

The persistent FFN parameter state is  $\Theta(3dm)$  elements per layer. A fully materialized rank- $r$  descriptor set is only  $\Theta(3r(d+m))$  elements, and a tiled kernel can reduce the live scratch working set further to  $\Theta(r(d_{\text{tile}} + m_{\text{tile}}))$ . These are storage-footprint and traffic estimates. They are not claims that every descriptor or activation is resident in on-chip SRAM at once; the no-write-back contract says that generated descriptor tiles should be consumed at the closest available scratch/cache level and not committed as durable off-chip model weights.

### 4.5 Approximation

Let  $\widetilde{W}_p = U_r^{(p)} V_r^{(p)}$  be the generated approximation of  $W_p$ . If  $\|W_p - \widetilde{W}_p\|_F \leq \epsilon_p$ , then for bounded activations and  $L_\phi$ -Lipschitz nonlinearity,

$$\|Y - \widetilde{Y}\|_F \leq C_1 L_\phi \|X\|_F (\epsilon_{\text{gate}} + \epsilon_{\text{up}}) + C_2 \|X\|_F \epsilon_{\text{down}}, \quad (13)$$

where  $C_1$  and  $C_2$  depend on activation bounds inside the FFN. This connects rank selection to output distortion, while leaving room for end-to-end distillation to improve beyond static SVD.

### 4.6 Eligibility and Benefit Routing

The systems advantage of an ephemeral descriptor is realized only on calls that can safely use it. We therefore treat MWG-EW as a budgeted decision problem rather than a blanket replacement. For a token or token group  $x$ , let  $\Delta(x) = \mathcal{L}_{\text{dense}}(x) - \mathcal{L}_{\text{mwg}}(x)$  be the measured benefit of using the generated path on training text. A deployable router learns  $\hat{\Delta}(x)$  from features available before the target FFN executes and patches the top-scoring budgeted subset:

$$g_{\mathcal{B}}(x) = \mathbf{1}\{\hat{\Delta}(x) \geq q_{1-\mathcal{B}}(\hat{\Delta})\}. \quad (14)$$

Table 2: Static SVD baseline on 18 pretrained FFN matrices. Low ranks are excellent for bytes but weak as fixed approximations, motivating learned conditional generation.

| Rank $r$ | Mean energy | Min energy | Mean cosine |
|----------|-------------|------------|-------------|
| 32       | 7.17%       | 4.27%      | 0.266       |
| 64       | 11.98%      | 8.16%      | 0.345       |
| 128      | 20.12%      | 15.46%     | 0.448       |
| 256      | 33.79%      | 28.65%     | 0.582       |
| 512      | 55.61%      | 50.75%     | 0.748       |
| 1024     | 85.13%      | 82.42%     | 0.925       |

The mixed FFN is

$$Y_{\text{mix}}(x) = (1 - g_{\mathcal{B}}(x))Y_{\text{dense}}(x) + g_{\mathcal{B}}(x)\tilde{Y}_{\text{mwg}}(x). \quad (15)$$

This objective is closer to systems deployment than an oracle frontier: the router observes hidden-state summaries and sequence metadata, not the evaluation loss of the dense and patched model. It also turns negative evidence into useful structure. If always-patched replacement fails but  $\Delta(x)$  is predictable for a small subset, then the method still yields a practical bandwidth budget.

**Proposition 2** (Budgeted value condition). *Let  $S_{\mathcal{B}} = \{x : g_{\mathcal{B}}(x) = 1\}$ . Suppose a full generated call saves  $s_r$  bytes or communication units relative to the dense call, and let  $o_r$  be the router, launch, staging, and any dense-residency overhead paid per candidate call. The expected net systems saving is bounded by*

$$\Pr[x \in S_{\mathcal{B}}]s_r - o_r. \quad (16)$$

*The replacement is quality-preserving whenever*

$$\mathbb{E}[\mathcal{L}_{\text{mwg}}(x) - \mathcal{L}_{\text{dense}}(x) \mid x \in S_{\mathcal{B}}] \leq 0. \quad (17)$$

*Thus value comes from the product of per-call descriptor savings and predictable eligibility, and the end-to-end saving can be much smaller than the full-call traffic ratio when the safe patch fraction is small.*

## 4.7 Static Low-Rank Is Not the Claim

To separate MWG-EW from ordinary compression, we analyze a static truncated-SVD baseline on 18 FFN matrices from a Qwen-1.5B-family checkpoint. Table 2 reports mean retained singular-value energy and reconstruction cosine. At ranks that are attractive for bandwidth, static SVD preserves too little FFN structure to be a complete solution. This is evidence for the generator side of MWG-EW: the low-rank factors should be conditional operands, not merely fixed compressed weights.

## 5 Systems Realization

The architectural claim is only meaningful if the runtime keeps the generated factors ephemeral. The current prototype measures the unfused associative path; the following contract describes the production implementation needed to validate the full no-write-back claim:

- **Generator placement.** The generator runs adjacent to the FFN kernel and emits descriptor tiles directly into scratch memory or the nearest cache-managed staging region.



Table 3: Positioning of MWG-EW relative to common efficient-Transformer mechanisms. The distinguishing axis is parameter lifetime: whether the weight-like object is durable model state or a volatile runtime operand.

| Method family      | Durable object    | Primary saving    | MWG-EW difference                    |
|--------------------|-------------------|-------------------|--------------------------------------|
| Quantization       | Quantized weights | Bits per weight   | Changes lifetime, not only precision |
| LoRA/adapters      | Low-rank factors  | Trainable state   | Factors are generated operands       |
| MoE                | Expert weights    | Activated compute | Expert weights remain persistent     |
| FlashAttention     | Activations       | Attention IO      | Targets FFN weight movement          |
| MLA/KV compression | Past-token state  | KV-cache memory   | Orthogonal to FFN descriptors        |

- **Associative execution.** The kernel computes  $XU_r$  followed by  $ZV_r$ , fusing activation, gate, and down-projection staging where the target accelerator permits it.
- **Eligibility control.** A cheap router decides whether the dense or ephemeral path should execute for the current token group. This router is part of the runtime path, not an offline oracle.
- **Lifecycle enforcement.** Descriptor buffers are marked volatile; instrumentation checks descriptor read/write traffic and verifies that generated factors do not reappear as persistent model tensors or optimizer state.

The intended kernel is a streaming descriptor pipeline. For each tile, the runtime generates or blends the required rank channels, performs the two associative matrix products, fuses the activation/gating operation, and then reuses the tile storage for the next token group. If the hardware lacks explicit software-managed SRAM, the same contract can be approximated at the nearest cache level, but the paper then reports descriptor traffic as an estimate until hardware counters confirm off-chip behavior.

This blueprint also clarifies why the current prototype is useful but not the final systems artifact. The benchmark measures the unfused associative path and shows that trainable-state and communication reductions can translate into latency improvements for full generated calls. It does not prove the final no-write-back traffic claim. A fused kernel and hardware counters are required to show that descriptor tiles avoid off-chip write-back in the intended runtime.

## 6 Experimental Protocol

**Execution setup.** The large-scale reliability run uses the ASI1 Huanxin environment with eight visible Ascend 910B2 NPUs. We evaluate one, two, four, and eight data-parallel ranks by setting `ASCEND_RT_VISIBLE_DEVICES` to the corresponding device set and launching PyTorch distributed execution with HCCL. Each configuration is repeated three times. The ASI2 replication uses the same geometry, launcher, and three-repeat policy, but with the ASI2 environment and a subset of ranks appropriate to the available device sets. The June 2026 ASI3 sweep uses all eight visible NPUs (`ASCEND_RT_VISIBLE_DEVICES=0,1,2,3,4,5,6,7`) and evaluates the wider

rank set  $r \in \{32, 64, 96, 128, 192, 256, 384, 512\}$ . The raw artifacts record platform metadata for reproducibility, but the method and results below are reported in terms of model geometry, rank, latency, state size, estimated traffic, and collective communication.

**Geometry.** The large settings use an 8B-scale FFN block with  $d = 4096$ ,  $m = 14336$ , decode shape  $B = 1, S = 128$ , and training shape  $B = 2, S = 512$ . The reliability/scaling sweep reported below uses representative ranks  $r \in \{64, 128, 256\}$  for the three-repeat reliability tables and the wider rank set for the fresh three-repeat ASI3 sweep.

**Validation path.** The benchmark uses a small smoke geometry  $(d, m) = (512, 1408)$  with short iteration counts before running the 8B-scale geometry. This keeps the expensive distributed sweep separate from basic checks of output writing, tensor shapes, and collective communication.

**Metrics.** We report forward latency, backward latency, throughput, trainable parameter bytes, estimated descriptor traffic, dense-to-MWG-EW speedup, all-reduce latency, and all-reduce bytes.

**Checkpoint quality and end-to-end probes.** To avoid treating systems speed as a proxy for model quality, we separately distill the FFN output of layer 0 from a local Qwen2.5-1.5B-Instruct checkpoint stored in the ASI1 file system. The teacher uses the checkpoint `gate_proj`, `up_proj`, and `down_proj` matrices with  $d = 1536$  and  $m = 8960$ . We compare static truncated SVD, a trainable persistent low-rank FFN, and the same low-rank factors with the MWG-EW rank-scale generator. The reported run uses eight NPUs, fp32 training for numerical stability, ranks 64 and 128, 16 short distillation steps, and four evaluation batches. We then run a stronger ASI3 quality path on the same checkpoint: text-activation distillation, expert-residual MWG-EW students, saved student checkpoints, and direct replacement of selected Qwen MLP modules followed by causal-language-model perplexity evaluation. This second probe is still small, but it is end-to-end: the patched model is evaluated through the full Transformer, not only by block MSE. We use this second path as the headline quality probe because causal-language-model perplexity is the quantity the paper ultimately cares about; the layer-0 block-MSE distillation is retained only as a controlled checkpoint-quality reference. Within that end-to-end path, layer 16 is selected as the best current single-layer candidate by the per-layer calibration sweep reported in Table 7, and all headline quality and routing results in Sections 7–8 use that layer.

**Reproducibility controls.** Held-out recovery, broad validation, and router launchers require explicit text paths and reject `DEFAULT_TEXTS` fallback through a `--require-texts` guard. The manifest-driven validation harness rejects empty or undersized corpora, and router splits record exact train/eval files, seeded partitions, and overlap removal. These controls keep the positive selective-routing result comparable across future replications and prevent the paper from mixing leakage-prone router gains with held-out broad evaluation.

**Deployable selective routing protocol.** The key algorithmic validation step is a non-oracle dense-fallback policy for the best current single-layer patch. The router labels train tokens by dense-vs-patched loss delta, fits a small ridge regressor from features available before the target FFN executes, and evaluates thresholded patch fractions with actual mixed dense/patch forward passes. The features are summary statistics of the layer-16 pre-FFN hidden state and sequence occupancy, not evaluation-set dense-vs-patched losses or post-hoc perplexity comparisons. This protocol tests MWG-EW as a selective runtime optimization rather than an indiscriminate layer replacement.

On 2026-06-10 and 2026-06-11 we also evaluated the calibrated layer-16 checkpoint in stricter ASI3 follow-ups: manifest-driven broad always-patched validation across GSM8K, MBPP, Alpaca-tail, Wikitext-103, and HellaSwag, a corrected hybrid manifest, and a disjoint router/global train/eval split with suite-balanced router fitting. These runs use explicit manifests and actual dense/patch or mixed forward passes.

**Artifact trail.** The reported systems run was executed on ASI1 and kept under the legacy run identifier `ASI3_reliability_scaling_20260518T072507Z`. It contains 12 raw JSON benchmark artifacts: three repeats each for world sizes 1, 2, 4, and 8, plus an aggregate JSON/Markdown summary. The local repository keeps the pulled summary, status file, and log under `results/pulled_asii1/`; the same experiment code and paper artifacts are mirrored to the project S3 folder. Historical ASI1/ASI3 systems artifacts and later ASI2 replications remain separable in the artifact names through explicit environment-label tagging. The June 2, 2026 ASI2 reliability sweep wrote nine raw JSON benchmark artifacts under the remote results directory. Here we use the verified per-run artifacts inspected through the ASI2 shell and limit the replication claim to those observed runs. The fresh June 5, 2026 ASI3 sweep was launched after restoring the project into `/vllm-workspace/mwg-ew-transformer-research`; its log records `nproc=8` and all eight visible device ids.

## 7 Numerical Evidence

**Evidence status.** The results below were measured on 2026-05-18 on ASI1 using all eight visible NPUs for the largest setting, `torch.distributed.run`, HCCL, and fp16 tensors. The benchmark uses the 8B-scale FFN geometry  $(d, m) = (4096, 14336)$ , decode shape  $(1, 128, 4096)$ , and training shape  $(2, 512, 4096)$ . Each row reports the mean and standard deviation over three independent launches. Each launch used three warmup iterations, ten forward timing iterations, three training iterations, and five all-reduce iterations. These measurements establish systems behavior for the associative low-rank path and distributed communication; language-model quality is evaluated separately with the checkpoint-backed gates and routers below.

**Interpretation.** The systems result is the clearest value signal. At eight ranks,  $r = 128$  reduces descriptor traffic from 336.0 MiB to 13.5 MiB, reduces trainable FFN state from 336.0 MiB to 18.254 MiB after generator overhead, improves mean training latency from  $6.4874 \pm 0.8606$  ms to  $4.2047 \pm 0.0299$  ms, and increases measured training throughput from 1.279M to 1.948M tokens/s. The fastest eight-rank point is  $r = 256$ , reaching  $1.572 \pm 0.165$ x training speedup and 1.992M tokens/s, while still reducing descriptor traffic by 12.4x and trainable FFN state by 10.3x. The lower-rank  $r = 64$  point gives the strongest compression, with 49.8x descriptor-traffic reduction and 30.2x trainable-state reduction. Across 2, 4, and 8 ranks, the all-reduce proxy drops from dense latencies of 19.66, 11.49, and 6.91 ms to low-rank latencies near 1–2.5 ms, showing that ephemeral descriptor state can translate directly into distributed communication savings. Tables 4 and 5 together establish that the directional systems gain holds across one, two, four, and eight devices and is not an artifact of a single world size.

The new ASI3 sweep in Table 18 strengthens that value signal because it tests a wider rank range on all eight NPUs with three independent launches. Dense training throughput is 1.483M tokens/s on average in this run family. Every tested generated-rank configuration improves training throughput, from 2.281M tokens/s at rank 32 to 2.440M tokens/s at rank 96, while preserving large descriptor-byte reductions. The best mean training speedup is rank 96: 33.2x descriptor-byte

Table 4: Reliability and scaling of 8B-scale FFN training on ASI1 across world sizes 1–8 and descriptor ranks 64–256. Columns report per-iteration training latency (ms), speedup relative to the rank-dropped (– rank) baseline at the same world size, and throughput in tokens per second. Each cell is mean $\pm$ std over three launches. The current prototype uses unfused associative execution; fused no-write-back kernels are the next systems step.

| World | Rank | Train ms            | Train speedup      | Train tok/s |
|-------|------|---------------------|--------------------|-------------|
| 1     | –    | 4.9742 $\pm$ 0.0239 | 1.000x             | 0.206M      |
| 1     | 64   | 3.3785 $\pm$ 0.1928 | 1.475 $\pm$ 0.075x | 0.304M      |
| 1     | 128  | 3.2545 $\pm$ 0.1694 | 1.531 $\pm$ 0.071x | 0.315M      |
| 1     | 256  | 3.2879 $\pm$ 0.1327 | 1.514 $\pm$ 0.055x | 0.312M      |
| 2     | –    | 5.8624 $\pm$ 0.7332 | 1.000x             | 0.353M      |
| 2     | 64   | 3.9907 $\pm$ 0.1104 | 1.472 $\pm$ 0.211x | 0.513M      |
| 2     | 128  | 4.7966 $\pm$ 1.5295 | 1.268 $\pm$ 0.214x | 0.453M      |
| 2     | 256  | 3.8896 $\pm$ 0.0662 | 1.507 $\pm$ 0.181x | 0.527M      |
| 4     | –    | 6.1351 $\pm$ 1.1496 | 1.000x             | 0.682M      |
| 4     | 64   | 4.1078 $\pm$ 0.1168 | 1.497 $\pm$ 0.303x | 0.998M      |
| 4     | 128  | 4.4547 $\pm$ 0.6583 | 1.412 $\pm$ 0.423x | 0.932M      |
| 4     | 256  | 4.1109 $\pm$ 0.0797 | 1.496 $\pm$ 0.312x | 0.997M      |
| 8     | –    | 6.4874 $\pm$ 0.8606 | 1.000x             | 1.279M      |
| 8     | 64   | 4.8044 $\pm$ 1.1034 | 1.373 $\pm$ 0.182x | 1.760M      |
| 8     | 128  | 4.2047 $\pm$ 0.0299 | 1.542 $\pm$ 0.196x | 1.948M      |
| 8     | 256  | 4.1222 $\pm$ 0.2432 | 1.572 $\pm$ 0.165x | 1.992M      |

Table 5: State size, descriptor traffic, and measured all-reduce latency for the same ASI1 reliability run. AllReduce latency cells are mean ms over three launches for the corresponding world size.

| Rank | Params MiB | Descriptor MiB | Traffic red. | AR 2    | AR 4    | AR 8   |
|------|------------|----------------|--------------|---------|---------|--------|
| –    | 336.000    | 336.000        | 1.000x       | 19.6627 | 11.4896 | 6.9101 |
| 64   | 11.127     | 6.750          | 49.778x      | 1.2599  | 1.1655  | 1.1549 |
| 128  | 18.254     | 13.500         | 24.889x      | 1.6772  | 1.4632  | 1.2786 |
| 256  | 32.507     | 27.000         | 12.444x      | 2.5398  | 1.8239  | 1.4182 |

reduction, 22.9x trainable-state reduction, and  $1.646 \pm 0.026x$  training speedup. Rank 128 remains a balanced operating point: 24.9x descriptor-byte reduction, 18.4x trainable-state reduction, and  $1.638 \pm 0.030x$  training speedup. All ranks from 32 through 512 remain above 1.5x mean training speedup, showing that the result is not a narrow rank accident. Forward speedup is variable and the unfused path is still not an inference-latency claim, so the paper treats the fresh ASI3 result as strong evidence for training and communication efficiency.

For a top-level systems claim, the ASI3 sweep is the most reliable current artifact because it combines full eight-NPU execution, three independent launches, and a broad rank range. The curve is not tuned to one point: ranks 32–512 all remain above 1.5x mean training speedup while descriptor-byte reduction ranges from 6.2x to 99.6x. This supports the paper’s central systems message that FFN parameter movement and synchronization can be reduced by changing parameter lifetime, even before a fused no-write-back kernel is built.

The ASI2 replication preserves the same direction on a new run family. Averaged over the

Table 6: Checkpoint-backed FFN approximation on ASI1 using Qwen2.5-1.5B layer 0. Lower relative MSE and higher cosine are better. The short eight-NPU distillation run improves over static SVD and motivates the selective routing protocol used for language-model validation.

| Method              | Rank | Rel. MSE | Cosine   | Params MiB | Traffic red. |
|---------------------|------|----------|----------|------------|--------------|
| Static SVD          | 64   | 0.989094 | 0.133395 | –          | 20.488x      |
| Persistent low-rank | 64   | 0.966727 | 0.225573 | 9.096      | 20.488x      |
| MWG-EW rank-scale   | 64   | 0.966702 | 0.225632 | 9.096      | 20.488x      |
| Static SVD          | 128  | 0.971749 | 0.187598 | –          | 10.244x      |
| Persistent low-rank | 128  | 0.942873 | 0.257047 | 17.066     | 10.244x      |
| MWG-EW rank-scale   | 128  | 0.942804 | 0.257052 | 17.066     | 10.244x      |

Table 7: Layer-16 calibration evidence on Qwen2.5-1.5B. Lower PPL ratio is better. The key comparison is the same 51,267-token held-out Belle slice: LM-CE calibration improves the MWG-EW layer-16 patch from the uncalibrated gate and outperforms the same-rank persistent low-rank control. The small diverse64 and heldout12 rows are auxiliary checks for layer eligibility.

| Patch policy                                    | Rank | Tokens | Dense PPL | Patched PPL / ratio |
|-------------------------------------------------|------|--------|-----------|---------------------|
| Layer 16 only, Belle256                         | 384  | 50838  | 8.37      | 10.77 / 1.287x      |
| Layer 16 persistent low-rank, Belle256          | 384  | 51267  | 8.44      | 11.59 / 1.373x      |
| Layer 16 only, Belle256, logit-MSE calibrated   | 384  | 51267  | 8.44      | 10.03 / 1.189x      |
| Layer 16 only, Belle256, LM-CE calibrated       | 384  | 51267  | 8.44      | 9.96 / 1.180x       |
| Layer 16 only, ASI3 2026-06-11 LM-CE calibrated | 384  | 17812  | 3.75      | 2.63 / 0.701x       |
| Layer 16 only, ASI3 2026-06-24 LM-CE calibrated | 384  | 35374  | 3.69      | 1.49 / 0.404x       |
| Layer 16 only, diverse64, LM-CE calibrated      | 384  | 2736   | 62.37     | 67.80 / 1.087x      |
| Layer 16 only, heldout12, LM-CE calibrated      | 384  | 1280   | 58.12     | 60.48 / 1.040x      |

nine ASI2 artifacts, mean training speedups are about 1.65x at world 1, 1.55x–1.59x at world 2, and 1.65x–1.81x at world 4, depending on rank. Mean forward speedups remain below 1.0x in every group, ranging roughly from 0.57x to 0.80x, so the result is a training and communication improvement rather than a decode latency win. The replication therefore strengthens the promise without changing the boundary.

The checkpoint-backed quality results identify the algorithmic source of value. Table 6 shows that static SVD alone is not competitive, and that FFN-output gains over a persistent low-rank control are small before routing. Table 7 then identifies layer 16 as the best current single-layer candidate and shows that LM-CE calibration improves the Belle held-out ratio to 1.180x, materially better than the same-rank persistent low-rank control at 1.373x. This is the setup that makes the selective-routing result possible. Table 8 shows why calibration matters: older always-patched broad controls were above dense, but the fresh calibrated ASI3 checkpoint improves the three-suite broad manifest to 0.8715x of dense, with the weakest suite still slightly below dense at 0.9995x.

Table 8: Aggregate broad controls for always-patched layer-16 replacement and durable same-rank low-rank baselines. These rows are included only to scope the deployment claim: older uncalibrated or weakly calibrated checkpoints were not broadly safe, while the fresh ASI3 calibrated checkpoint is positive on the three-suite broad manifest. The five-suite row adds Wikitext-103 and HellaSwag and is positive in aggregate, although not every added suite improves. Lower PPL ratio is better.

| Policy            |                  |                                     | Tokens | Weighted ratio | Max suite |
|-------------------|------------------|-------------------------------------|--------|----------------|-----------|
| MWG               | LM-CE            | always-patched, older broad control | 91,352 | 1.2464x        | –         |
| Persistent        | low-rank         | always-patched                      | 66,873 | 1.2795x        | –         |
| Persistent        | low-rank + LM-CE | always-patched                      | 91,352 | 1.3099x        | –         |
| ASI3              | 2026-06-10       | calibrated                          | 47,608 | 0.8715x        | 0.9995x   |
| MWG, three suites |                  |                                     |        |                |           |
| ASI3              | 2026-06-11       | calibrated                          | 74,456 | 0.9352x        | 1.1284x   |
| MWG, five suites  |                  |                                     |        |                |           |

Table 9: Three-fold token-level router cross-validation over the pooled router corpus. The 512 unique router lines are repartitioned into explicit 256/256 train/eval folds, then evaluated with actual mixed dense/patch forward passes. This tests split sensitivity within the router distribution; it is not a replacement for independent broad benchmarks. Lower PPL ratio is better.

| Target patch frac. | Mean actual patch frac. | Mean ratio | Min ratio | Max ratio |
|--------------------|-------------------------|------------|-----------|-----------|
| 0.01               | 0.0009                  | 0.9989x    | 0.9955x   | 1.0009x   |
| 0.03               | 0.0038                  | 0.9940x    | 0.9870x   | 1.0009x   |
| 0.05               | 0.0075                  | 0.9896x    | 0.9796x   | 1.0007x   |
| 0.10               | 0.0193                  | 0.9787x    | 0.9644x   | 0.9974x   |
| 0.25               | 0.0695                  | 0.9462x    | 0.9282x   | 0.9618x   |
| 0.50               | 0.1766                  | 0.9218x    | 0.9137x   | 0.9373x   |

The ASI3 layer-16 recovery runs add a sharper generator-side signal at the same rank. The strongest recovered MWG-EW checkpoint reaches relative MSE 0.101849 and cosine 0.947711, while the static SVD baseline stays at 0.836682 and 0.404468. A later two-NPU ASI3 rerun independently preserves the same separation, with MWG-EW at relative MSE 0.130428 and cosine 0.932599 versus static SVD at 0.835157 and 0.406409. That is not a blanket systems result, but it is a meaningful algorithmic gain because the same rank family now supports a much better conditional descriptor fit than a fixed truncated decomposition.

The completed 2026-06-11 ASI3 rerun also gives a strong calibration-slice language-model result: after LM-CE calibration on explicit text, dense PPL is 3.748 and patched PPL is 2.628, a 0.701x ratio. We use this as evidence that the patch can become language-model beneficial under explicit calibration, not as a replacement for leakage-clean broad evidence. The three-suite ASI3 broad validation removes that ambiguity for GSM8K, MBPP, and Alpaca-tail: GSM8K improves to 0.801x, MBPP to 0.763x, and Alpaca-tail to 0.9995x, for a token-weighted ratio of 0.8715x. The corrected five-suite ASI3 validation then adds Wikitext-103 and HellaSwag and remains positive in aggregate at 0.9352x over 74,456 tokens. Wikitext-103 and HellaSwag individually regress under

Table 10: Fresh ASI3 2026-06-10 disjoint router/global validation for the calibrated layer-16 checkpoint. The always-patched row evaluates the full patch on the eval split; router rows use a dense-fallback policy trained on the disjoint train split and executed with actual mixed dense/patch forward passes. Lower PPL ratio is better.

| Policy             | Actual patch frac. | Tokens | PPL ratio | Max suite ratio |
|--------------------|--------------------|--------|-----------|-----------------|
| Always patched     | 1.0000             | 45,027 | 0.8605x   | 0.9874x         |
| Router target 0.05 | 0.0072             | 45,384 | 0.9851x   | –               |
| Router target 0.10 | 0.0192             | 45,384 | 0.9607x   | –               |
| Router target 0.25 | 0.0688             | 45,384 | 0.9127x   | –               |

Table 11: Corrected ASI3 2026-06-11 five-suite validation. The always-patched token-router row aggregates GSM8K, MBPP, Alpaca-tail, Wikitext-103, and HellaSwag. The hybrid row is a corrected manifest evaluation and must be read separately because it uses a different token count. Lower PPL ratio is better.

| Policy                     | Actual patch frac. | Tokens | PPL ratio | Note               |
|----------------------------|--------------------|--------|-----------|--------------------|
| Always patched, five-suite | 1.0000             | 74,456 | 0.9352x   | aggregate positive |
| Router target 0.05         | 0.0359             | 75,073 | 0.9692x   | mixed forward      |
| Router target 0.10         | 0.0669             | 75,073 | 0.9552x   | mixed forward      |
| Router target 0.25         | 0.1471             | 75,073 | 0.9403x   | mixed forward      |
| Corrected hybrid manifest  | 1.0000             | 39,394 | 0.8493x   | separate corpus    |

the always-patched policy, so the defensible claim is aggregate five-suite benefit plus strong positive transfer on GSM8K and MBPP, not uniform suite-wise improvement.

The broad router sweeps are the strongest current quality evidence. The router uses token CE deltas on explicit train text, features available before the FFN executes, and actual mixed dense/patch forward passes at evaluation time. In three-fold in-distribution validation (Table 9), mean mixed-forward ratios reach 0.946x at 6.9% actual patching and 0.922x at 17.7% actual patching. In the leakage-clean broad setting (Table 14), token-weighted ratios are below dense at 2.1% and 8.5% actual patching, but the max-suite ratio at target 0.25 is still above dense. The three-seed broad sweep in Table 15 improves over dense at target fractions 0.05, 0.10, and 0.25 on every seed, with actual patch fractions 0.49%, 1.82%, and 8.52%; its max seed-level broad ratios are 0.9997x, 0.9952x, and 0.9929x, respectively. The stricter leave-suite-out sweep (Table 16) keeps only targets 0.05 and 0.10 below dense on every seed, with actual patch fractions 2.84% and 5.65%, mean ratios of 0.997x and 0.993x, and max seed-level broad ratios of 0.9996x and 0.9943x. Target 0.25 remains promising but not fully robust there: its mean ratio stays below dense at 0.994x, but the max seed-level broad ratio rises slightly above dense to 1.0004x. This is the deployable positive result and its boundary: a learned dense-fallback router can patch a small, measurable fraction of tokens while preserving or improving perplexity, but the wider patch regime is still fragile. It is not evidence that the full-call systems savings apply to every FFN invocation, and it is currently validated only on three held-out suite families: math reasoning, code generation, and instruction following.

The fresh ASI3 router/global split in Table 10 makes that deployment story stronger. On a disjoint 357-example eval split, the full calibrated layer-16 patch reaches a 0.8605x PPL ratio. The same checkpoint under a learned dense-fallback router improves monotonically across the tested

Table 12: Fresh 2026-06-24 ASI3 three-seed (seeds 0, 1, 2) five-suite token-router replication for an independently retrained and LM-CE-calibrated layer-16 checkpoint, under leakage-clean seeded splits with balanced-ridge fitting and fail-on-overlap controls. The always-patched per-suite rows are deterministic given the checkpoint, so seed variance is negligible. This run is more conservative than the single-run 2026-06-11 five-suite number: the always-patched five-suite aggregate is above dense, driven by the out-of-family suites, even though the same checkpoint is strongly positive in-distribution (0.404x calibration slice) and on GSM8K. Lower PPL ratio is better.

| Always-patched suite               | Mean PPL ratio | Seeds |
|------------------------------------|----------------|-------|
| GSM8K                              | 0.9652x        | 3     |
| MBPP                               | 1.2162x        | 3     |
| Alpaca-tail                        | 1.2398x        | 3     |
| Wikitext-103                       | 1.7973x        | 3     |
| HellaSwag                          | 2.5485x        | 3     |
| Five-suite aggregate               | 1.4086x        | 3     |
| Router target 0.05 (actual 0.6229) | 1.1725x        | 3     |
| Router target 0.10 (actual 0.8169) | 1.3424x        | 3     |
| Router target 0.25 (actual 1.0000) | 1.4117x        | 3     |

budgets: 0.9851x at 0.72% actual patching, 0.9607x at 1.92%, and 0.9127x at 6.88%. The gain is therefore not only an oracle or calibration-slice effect; it survives an explicit train/eval split and actual mixed forwards. The paper’s claim remains bounded because the experiment patches one FFN layer, but the algorithmic result is now positive in both full-patch and selective-patch modes for the calibrated checkpoint.

The corrected 2026-06-11 ASI3 five-suite run in Table 11 is the broadest current validation. The always-patched aggregate remains below dense at 0.9352x over 74,456 tokens, and the selective router preserves positive aggregate ratios up to the tested 14.71% actual patch-token budget. The corrected hybrid manifest is stronger still at 0.8493x over 39,394 tokens. This improves the paper’s breadth, while also clarifying the next research target: suite-robust eligibility control is needed because Wikitext-103 and HellaSwag do not yet improve under unconditional layer replacement.

A fresh 2026-06-24 ASI3 reproduction sharpens this boundary rather than softening it. We independently retrained and LM-CE-calibrated the layer-16 checkpoint and re-ran the five-suite token-router evaluation across three seeds under leakage-clean seeded splits, balanced-ridge fitting, and fail-on-overlap controls (Table 12). In-distribution the checkpoint is strongly positive, improving the calibration slice from dense PPL 3.69 to 1.49 (0.404x over 35,374 tokens), and it remains positive on GSM8K at 0.965x. However, the always-patched five-suite aggregate is above dense at 1.409x with negligible seed variance, driven by the out-of-family suites (HellaSwag 2.549x, Wikitext-103 1.797x), and the suite-local router threshold over-selects (actual patch fractions of 62%, 82%, and 100% for targets 5%, 10%, and 25%), so its frontier stays above dense at 1.172x, 1.342x, and 1.412x. We report this conservative replication explicitly: the defensible positive claims are the in-distribution calibration benefit and GSM8K transfer, while a uniformly safe five-suite always-patched policy is *not* established under three-seed leakage-clean evaluation, and a tighter eligibility policy than suite-local thresholding is required to recover a net selective win on the broad mix.

Table 13 summarizes the causal ladder behind the result. The gain is not explained by fixed low-rank compression: static SVD at the same rank has weak reconstruction fidelity. It is also not merely a persistent low-rank effect: the same-rank durable baseline is worse on Belle held-out text.



Table 13: Algorithmic ablation ladder for the layer-16 Qwen2.5-1.5B patch. The table separates fixed compression, learned low-rank controls, calibration, broad validation, and deployable routing. Lower PPL ratio is better.

| Variant                        | Rank | Evaluation          | PPL ratio / fit              |
|--------------------------------|------|---------------------|------------------------------|
| Static SVD                     | 384  | FFN reconstruction  | rel. MSE 0.8367, cos. 0.4045 |
| MWG-EW expert residual         | 384  | FFN reconstruction  | rel. MSE 0.1018, cos. 0.9477 |
| Persistent low-rank            | 384  | Belle held-out      | 1.373x                       |
| MWG-EW LM-CE calibrated        | 384  | Belle held-out      | 1.180x                       |
| MWG-EW LM-CE calibrated        | 384  | Broad manifest      | 0.8715x                      |
| MWG-EW + dense-fallback router | 384  | Router/global eval  | 0.9127x at 6.88% patched     |
| MWG-EW LM-CE calibrated        | 384  | Five-suite manifest | 0.9352x over 74,456 tokens   |
| MWG-EW corrected hybrid        | 384  | Five-source hybrid  | 0.8493x over 39,394 tokens   |

Table 14: Leakage-clean broad token-router validation. For each broad suite, the router train pool removes exact lines overlapping that suite’s eval file before training. Values aggregate GSM8K, MBPP, and Alpaca-tail using token-weighted losses. Lower PPL ratio is better.

| Target patch frac. | Actual patch frac. | Token-weighted ratio | Min suite | Max suite |
|--------------------|--------------------|----------------------|-----------|-----------|
| 0.01               | 0.0005             | 1.0010x              | 1.0001x   | 1.0050x   |
| 0.03               | 0.0019             | 1.0008x              | 1.0005x   | 1.0025x   |
| 0.05               | 0.0055             | 0.9989x              | 0.9970x   | 1.0004x   |
| 0.10               | 0.0208             | 0.9916x              | 0.9648x   | 0.9996x   |
| 0.25               | 0.0849             | 0.9868x              | 0.9197x   | 1.0038x   |
| 0.50               | 0.1975             | 1.0140x              | 0.8587x   | 1.0498x   |

The positive regime appears after conditional generation, LM-CE calibration, and eligibility control are combined. This is the core algorithmic contribution that a reviewer should judge independently from the still-unfused systems prototype.

The implication of Table 19 is that the full-call descriptor-byte reduction cannot be naively multiplied by the patch fraction to obtain an end-to-end routed saving. At the small actual patch fractions where the selective router is quality-preserving (0.49% and 1.82% in the broad seed sweep), the effective routed descriptor bytes are only 12.2% and 45.3% of dense—a real reduction, but far smaller than the 24.9x per-call figure. Under the stricter leave-suite-out sweep, the target 0.10 row is *above* dense (140.6%), illustrating why the paper scopes its systems claim to the full-call and training settings and treats end-to-end routed savings as an open target.

Table 17 records the remaining systems instrumentation scope. On an ASI3 host with visible Ascend 910B2 NPUs and available `npu-smi/msprof`, the PyTorch profiler exposed CPU activities only and observed no fused-like MWG operator. The paper therefore reports descriptor traffic as a model-side estimate paired with measured ASI1 latency and all-reduce behavior, not as measured off-chip traffic. Off-chip counter confirmation is a required future fused-kernel experiment.

Table 15: Three-seed leakage-clean broad token-router sweep. Each seed uses 75% of the overlap-clean router train candidates for each broad suite, then evaluates actual mixed dense/patch forwards on GSM8K, MBPP, and Alpaca-tail. Values report the mean across seeds; min and max are seed-level token-weighted broad ratios. Lower PPL ratio is better.

| Target patch frac. | Mean actual patch frac. | Mean ratio | Min seed | Max seed |
|--------------------|-------------------------|------------|----------|----------|
| 0.01               | 0.0007                  | 1.0011x    | 1.0008x  | 1.0013x  |
| 0.03               | 0.0021                  | 0.9998x    | 0.9980x  | 1.0013x  |
| 0.05               | 0.0049                  | 0.9977x    | 0.9941x  | 0.9997x  |
| 0.10               | 0.0182                  | 0.9920x    | 0.9890x  | 0.9952x  |
| 0.25               | 0.0852                  | 0.9864x    | 0.9820x  | 0.9929x  |
| 0.50               | 0.2168                  | 1.0164x    | 0.9943x  | 1.0312x  |

Table 16: Leave-suite-out broad auxiliary token-router sweep. For each target broad suite, router training uses the explicit router texts plus the other two broad suites, with exact target-suite eval lines removed. Values report the mean across three split seeds; min and max are seed-level token-weighted broad ratios. Lower PPL ratio is better.

| Target patch frac. | Mean actual patch frac. | Mean ratio | Min seed | Max seed |
|--------------------|-------------------------|------------|----------|----------|
| 0.01               | 0.0043                  | 1.0007x    | 1.0004x  | 1.0009x  |
| 0.03               | 0.0166                  | 0.9989x    | 0.9980x  | 1.0004x  |
| 0.05               | 0.0284                  | 0.9975x    | 0.9964x  | 0.9996x  |
| 0.10               | 0.0565                  | 0.9926x    | 0.9914x  | 0.9943x  |
| 0.25               | 0.1380                  | 0.9943x    | 0.9889x  | 1.0004x  |
| 0.50               | 0.2806                  | 1.0274x    | 1.0076x  | 1.0488x  |

## 8 Discussion

MWG-EW is best understood as a bridge between neural architecture and runtime design. The neural component makes a low-rank conditional descriptor useful on some inputs; the systems component ensures that this descriptor does not quietly become another persistent tensor. Either side alone is insufficient. Static low-rank compression gives attractive byte counts but weak quality, while an unfused runtime can expose the right trend without proving the final no-write-back traffic claim.

The most important conceptual shift is from compression to *lifetime allocation*. A dense FFN weight is a long-lived object. A LoRA factor is smaller but still long-lived. An ephemeral descriptor is short-lived and context-bound. This gives model designers a new budget: not merely how many weights exist, but which weight-like objects deserve durability. For very large models, that budget is directly tied to memory bandwidth, optimizer state, checkpoint traffic, and collective communication.

The quality results suggest a concrete algorithmic agenda. The strongest current algorithm is not “replace every FFN,” but “learn the eligibility frontier.” Better benefit predictors, suite-local thresholds, layer-specific calibration, rank-adaptive descriptors, and multi-layer budget allocation can increase the fraction of FFN calls that safely use generated operands. In this view, the router is not an auxiliary patch; it is the mechanism that turns a conditional systems primitive into a deployable model component. The positive result is already useful because it converts a failing always-patched replacement into a verified selective operating regime.

Table 17: ASI3 runtime profiler probe for the current unfused prototype. The environment reported 8 visible Ascend 910B2 NPUs and available `npu-smi/msprof`, but this PyTorch-profiler capture exposed CPU activities only. It is therefore operator-level evidence, not full off-chip hardware-counter validation. Lower time is better.

| Case             | Descriptor MiB | Mean ms | Matmul-like ops | Fused-like op |
|------------------|----------------|---------|-----------------|---------------|
| Dense            | 64.50          | 0.3391  | 27              | no            |
| MWG-EW $r = 128$ | 5.53           | 0.7638  | 54              | no            |
| MWG-EW $r = 256$ | 11.06          | 0.7559  | 54              | no            |

Table 18: Fresh ASI3 three-repeat eight-NPU rank sweep on 2026-06-05. The run used `ASCEND_RT_VISIBLE_DEVICES=0,1,2,3,4,5,6,7`, world size 8, and the 8B-scale FFN geometry. Training speedup is mean $\pm$ std over three launches against the dense FFN in the same launch family; traffic reduction is the dense descriptor byte estimate divided by the rank- $r$  descriptor byte estimate.

| Rank | Params MiB | Traffic red. | Train ms          | Train speedup      | Train tok/s |
|------|------------|--------------|-------------------|--------------------|-------------|
| 32   | 7.564      | 99.556x      | $3.595 \pm 0.137$ | $1.538 \pm 0.026x$ | 2.281M      |
| 64   | 11.127     | 49.778x      | $3.409 \pm 0.100$ | $1.622 \pm 0.059x$ | 2.405M      |
| 96   | 14.691     | 33.185x      | $3.359 \pm 0.069$ | $1.646 \pm 0.026x$ | 2.440M      |
| 128  | 18.254     | 24.889x      | $3.375 \pm 0.069$ | $1.638 \pm 0.030x$ | 2.428M      |
| 192  | 25.380     | 16.593x      | $3.481 \pm 0.055$ | $1.589 \pm 0.067x$ | 2.354M      |
| 256  | 32.507     | 12.444x      | $3.429 \pm 0.083$ | $1.612 \pm 0.029x$ | 2.390M      |
| 384  | 46.760     | 8.296x       | $3.379 \pm 0.069$ | $1.636 \pm 0.029x$ | 2.425M      |
| 512  | 61.013     | 6.222x       | $3.414 \pm 0.117$ | $1.619 \pm 0.004x$ | 2.402M      |

**Relation to KV-cache compression and MLA.** The KV-cache memory used during autoregressive decoding is a different object from the FFN weight traffic studied here. A standard attention layer stores past keys and values with leading size  $\Theta(2Ln_hd_h)$  elements for sequence length  $L$ , head count  $n_h$ , and head dimension  $d_h$ ; in practical long-context serving this footprint is usually an off-chip memory allocation, although recently accessed tiles may pass through on-chip caches during an attention kernel. MLA reduces this attention-side storage by caching a latent key-value vector and reconstructing or absorbing key/value projections at attention time [5]. MWG-EW is orthogonal: it targets FFN projection weights, does not by itself compress the attention KV cache, and uses low-rank generation to change the lifecycle of weight operands rather than the representation of past-token activations.

**Current scope and submission readiness.** The present evidence supports a positive systems-and-routing claim suitable for a focused architecture or systems submission. The paper verifies four linked claims: generated descriptors can substantially reduce trainable FFN state and communication, conditional generation beats static SVD at the same rank, LM-CE calibration can make the best single-layer patch beneficial on broad perplexity, and a deployable dense-fallback router preserves additional gains at low actual patch fractions. Descriptor lifecycle claims still require low-level hardware performance counters for off-chip read and write transactions. In this revision, ASI2 execution produced a verified nine-run reliability artifact set, ASI3 produced a fresh three-repeat all-eight-NPU sweep over ranks 32–512, the corrected 2026-06-11 ASI3 validation produced a positive five-suite aggregate PPL result for the calibrated layer-16 checkpoint, and a fresh 2026-06-24 three-

Table 19: Effective descriptor-byte reduction implied by the quality-preserving router window. Full-call traffic reduction is the per-routed-call estimate from Table 5; effective routed descriptor bytes multiply that estimate by the observed actual patched-token fraction. Values above 100% mean the routed subset would move more descriptor bytes than one dense full layer call, although not more than replacing every call with generated descriptors. This table prevents confusing kernel level savings with end-to-end routed savings.

| Validation setting | Target | Actual patch frac. | Full-call red. | Effective routed bytes |
|--------------------|--------|--------------------|----------------|------------------------|
| Broad seed sweep   | 0.05   | 0.49%              | 24.9x          | 12.2% of dense         |
| Broad seed sweep   | 0.10   | 1.82%              | 24.9x          | 45.3% of dense         |
| Leave-suite-out    | 0.05   | 2.84%              | 24.9x          | 70.7% of dense         |
| Leave-suite-out    | 0.10   | 5.65%              | 24.9x          | 140.6% of dense        |

seed leakage-clean reproduction confirmed strong in-distribution and GSM8K benefit while showing that the always-patched five-suite aggregate is not yet uniformly safe under seeded splits. The main claim boundary is therefore precise: the paper establishes MWG-EW as a parameter-lifetime and selective execution contribution, not yet as a fused decode-latency win or a drop-in full-Transformer replacement. The highest-value follow-up experiments are suite-robust eligibility policies that recover a net five-suite selective win, fused-kernel hardware-counter validation, and multi-layer budget allocation.

**Limitations.** We state the boundary of the present evidence explicitly. First, the systems measurements use an *unfused* associative factor-matmul path; the no-write-back descriptor lifecycle is a specified runtime contract, not yet a measured off-chip behavior, because the available profiler exposed CPU activities only and observed no fused-like operator. Second, the validated systems claim is scoped to training throughput, trainable state, and distributed communication; the prototype is not faster on the unfused forward path, so no decode-latency claim is made. Third, the language-model quality evidence is anchored on a single calibrated FFN layer (layer 16) of one checkpoint family (Qwen2.5-1.5B); multi-layer and multi-model generalization is future work. Fourth, always-patched replacement is broadly safe only for the calibrated layer-16 checkpoint and only in aggregate over the five-suite manifest, with Wikitext-103 and HellaSwag individually regressing; the robust positive result is the *selective*, budgeted router regime at small actual patch fractions. Fifth, the broad held-out manifest currently covers math reasoning, code generation, and instruction following plus two added general/ commonsense suites, but does not yet cover dialogue or long-context families, so we do not make a broad-family deployment claim. We report all of these as boundary evidence rather than omitting the negative controls.

**Reproducibility.** Every reported number is tied to a named artifact. The systems sweeps are kept under explicit run identifiers (ASI3\_reliability\_scaling\_20260518T072507Z, the June 2026 ASI3 all-eight-NPU sweep, and the nine-run ASI2 reliability set), each recording platform metadata, device ids, and per-launch JSON. Quality and routing runs use explicit train/eval text paths, a `--require-texts` guard that rejects DEFAULT\_TEXTS fallback, manifest-driven corpora with minimum-size and overlap checks, and seeded train/eval partitions. The benchmark, distillation, calibration, router, and broad-validation scripts, the held-out manifest shapes, and the pulled result summaries are mirrored to a project artifact store. We report descriptor traffic as a model-side analytic estimate derived from the measured FFN geometry and clearly separate it from measured latency and all-reduce numbers in the evidence map (Table 1).

## 9 Conclusion

This paper argues that Transformer FFN weights can be treated not only as stored parameters, but as runtime-generated operands with explicit lifetimes. MWG-EW realizes this idea with conditional rank- $r$  descriptors, an associative execution path, and a proposed no-write-back lifecycle. Full generated calls reduce estimated descriptor traffic, trainable FFN state, and distributed gradient communication in an eight-rank 8B-scale FFN systems benchmark.

The checkpoint-backed Qwen2.5-1.5B experiments identify a concrete quality-preserving regime: layer 16 is the best current single-layer candidate, LM-CE calibration improves it relative to persistent low-rank control, and a benefit-supervised dense-fallback router preserves or improves broad perplexity at small actual patch-token fractions across leakage-clean, leave-suite-out, and corrected five-suite validation. The 2026-06-24 three-seed leakage-clean replication bounds this regime honestly: the same checkpoint is strongly positive in-distribution and on GSM8K, but the always-patched five-suite aggregate is above dense, so a uniformly safe always-patched policy is not yet established. The defensible contribution is therefore the *selective*, budgeted regime at small actual patch fractions, not blanket layer replacement. The broader contribution is the parameter-lifetime view itself. The paper therefore establishes a reproducible path for co-designing memory movement, synchronization, and model quality rather than optimizing them in isolation.

The new ASI2 reliability replication and fresh ASI3 three-repeat all-NPU sweep strengthen that path operationally: the same rank family scales repeatably across one, two, four, and eight-device settings, and the widest ASI3 rank sweep shows a stable 1.5x–1.65x training-side improvement across a broad rank range. It still does not license stronger decode claims. The highest-value follow-up experiments are suite-robust eligibility policies that recover a net five-suite selective win, a fused no-write-back kernel with hardware-counter confirmation of off-chip descriptor traffic, and multi-layer budget allocation across the FFN stack. Each of these targets a specific boundary stated in the evidence map, and each is required before the parameter-lifetime view can be turned into a deployable full-Transformer execution primitive.

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